

Cooperative Multi-Drone Routing under Deadlines and Battery Constraints via Attention-Based Multi-Agent Reinforcement Learning

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Abstract

This paper studies the cooperative routing of a fleet of M battery-constrained unmanned aerial vehicles (UAVs) serving a shared set of N customers with individual service deadlines, in a map that contains a small number of en-route chargers and a common depot. The problem is formulated as a sequential-turn multi-agent Markov decision process in which exactly one drone acts per global step and the team reward is the sum of per-drone contributions minus a penalty for any battery-depletion event. Four heuristic baselines are constructed to cover the main classes of fleet partitioning (spatial clustering, polar sweep, auction, and global earliest-deadline-first dispatch), and four multi-agent reinforcement learning agents are evaluated: independent proximal policy optimisation (I-PPO), asynchronous advantage actor-critic (A3C), multi-agent PPO (MAPPO) with a centralised critic, and an attention-based multi-agent policy that shares a cross-attention encoder across drones and builds a per-drone query token from the acting drone’s battery, time, and position. Experiments on a scenario grid over fleet size, customer count, customer spatial distribution, and deadline tightness show that the attention-based policy is the only method that is permutation-invariant in both the drone set and the customer set, and that it closes the load-imbalance gap that clustering heuristics leave on non-uniform instances. The findings indicate that architectural invariance across both axes is a prerequisite for deep multi-agent reinforcement learning to outperform well-tuned heuristics on this class of deadline- and battery-constrained routing problems.

Keywords: UAV fleet routing; multi-agent reinforcement learning; cross-attention; proximal policy optimisation; deadline-constrained vehicle routing; battery-aware scheduling.

1 Introduction

Time-critical aerial missions — search-and-rescue, emergency medical-supply delivery, and last-mile logistics under peak demand — increasingly rely on *fleets* of small unmanned aerial vehicles (UAVs) rather than a single vehicle [Mayer et al., 2019, Murphy, 2016]. The move from one drone to many is driven by two factors. First, the service area in a realistic disaster or delivery setting is too large for any single battery-constrained vehicle to cover

before deadlines expire [Galkin et al., 2019]. Second, commodity quadrotors have become inexpensive enough that deploying a coordinated fleet is operationally feasible, provided the dispatcher can decide, in real time, which drone serves which customer and when each drone diverts to a shared charger.

Multi-drone cooperative routing inherits the hard constraints of the single-drone problem — finite battery, individual service deadlines, and a small number of en-route chargers — and adds two cooperative decisions that no heuristic designed for a single vehicle can make well. The dispatcher must partition work across drones so that no drone is overloaded while others idle (load balancing), and it must resolve contention for chargers and for customers that lie in shared overlap regions (coordination). These two decisions interact: an aggressive partitioning that minimises total distance can strand a drone at the edge of the map when the charger queue fills up, while a partition that holds reserve capacity per drone wastes battery on empty returns. Heuristics drawn from the vehicle-routing literature — K-means clustering [Solomon, 1987], polar sweep [Gillett and Miller, 1974], iterative auctions [Bertsekas, 1988], and global earliest-deadline-first dispatch — each answer the partitioning question in a different way, but all four rely on a fixed offline assignment and cannot re-plan if a drone detours to a charger mid-mission.

Deep reinforcement learning offers a natural remedy because a trained policy conditions on the live state of every drone and every customer, and a single forward pass chooses the next action in microseconds [Bengio et al., 2021]. The first wave of combinatorial reinforcement learning used flat multi-layer perceptrons whose permutation-variance makes them brittle once the instance size moves away from training; the second wave, following Kool et al. [2019], adopted attention encoders that are permutation-invariant across the customer set and generalise cleanly across node counts. Attention has since become the architectural backbone of modern neural routing. Yet essentially all of this work targets the *single*-vehicle case. Extending cross-attention to a fleet requires a second invariance — over the drone set — and a training protocol that delivers a team reward under sequential-turn decisions. Existing multi-agent reinforcement learning (MARL) approaches, such as MAPPO [Yu et al., 2022], QMIX [Rashid et al., 2018], and value decomposition [Sunehag et al., 2018], provide useful training infrastructure but inherit the flat-MLP actor of the first wave and therefore do not give a permutation-invariant policy over the drone set unless the architecture is

modified.

This paper proposes an attention-based multi-agent reinforcement learning agent for cooperative multi-drone routing under deadlines and battery constraints. A shared cross-attention encoder processes the full set of nodes (depot, customers, chargers) into contextual embeddings, and the acting drone is represented by a learned query token built from its battery, its position, and an active-drone one-hot identifier. The query attends over the shared node embeddings; the resulting scores, after applying a feasibility mask for infeasible moves, are the action logits. A team critic pools the drone embeddings with a masked mean, giving a single scalar state value for the whole fleet. Training uses proximal policy optimisation [Schulman et al., 2017] with shared weights across drones, so that the sample complexity grows with the number of *transitions* rather than the product of drones and customers. The contributions of this work are as follows:

- Formulating cooperative multi-drone routing under deadline and battery constraints as a sequential-turn multi-agent Markov decision process with an action-feasibility mask and a team-additive reward, including a charger model that enforces battery and time feasibility at each step.
- Proposing an attention-based multi-agent policy in which a shared cross-attention encoder supplies permutation-invariance over the customer set and a per-drone query token supplies permutation-invariance over the drone set.
- Benchmarking four heuristic baselines (cluster, polar-sweep, auction, and global earliest-deadline-first) and three MARL baselines (I-PPO, A3C, MAPPO) on a scenario grid over fleet size, customer count, spatial distribution, and deadline tightness.
- Demonstrating scalability along both axes of the problem: the trained policy transfers to fleet sizes and customer counts not seen during training, with load-imbalance that remains flat as the fleet grows.

The remainder of the paper is organised as follows. Section 2 surveys multi-agent reinforcement learning and attention-based combinatorial policies and positions this work against both. Section 3 formalises the cooperative multi-drone routing problem as a sequential-turn multi-agent Markov decision process. Section 4 describes the attention-based multi-agent policy architecture and the training pipeline. Section 5 presents the

heuristic and MARL baselines, the scenario grid, and the evaluation protocol. Section 6 reports the main results. Section 7 discusses scalability, load balancing, and failure modes. Section 8 concludes and outlines extensions to heterogeneous fleets and communication-augmented variants.

2 Related Work

The literature on drone-fleet routing draws on three streams: classical vehicle routing heuristics adapted to aerial fleets, reinforcement-learning approaches to combinatorial routing, and multi-agent reinforcement learning for cooperative decision-making.

2.1 Heuristics for multi-vehicle routing

The vehicle-routing problem with time windows is NP-hard [Solomon, 1987, Vansteenwegen et al., 2011], so practical multi-vehicle solvers have long relied on cluster-first-route-second heuristics. Classical fleet partitioners include the K-means cluster-and-route family [Fisher and Jaikumar, 1981], polar sweep dispatching [Gillett and Miller, 1974], and iterative marginal-cost auctions [Bertsekas, 1988]. Each partitioner answers the load-balancing question offline: once a customer is assigned to a drone, it is not reassigned even if the drone detours to a charger mid-mission. For battery-constrained UAV fleets this static-assignment limitation has been documented by Dorling et al. [2017] and Galkin et al. [2019], who show that the total-distance optimum under capacity constraints is often far from the deadline-feasible optimum.

2.2 Attention-based policies for single-vehicle routing

Kool et al. [2019] introduced a cross-attention encoder-decoder policy for the travelling-salesman problem and the capacitated vehicle-routing problem, trained with REINFORCE under a rollout baseline. The attention mechanism delivers permutation-invariance across the customer set, which in turn enables the same weights to be used for different problem sizes. Extensions to deadline-constrained variants [Fan et al., 2023, Ma et al., 2023] show that a feasibility mask over the decoder logits is sufficient to enforce hard constraints when training is stable. The architecture used in this paper follows Kool’s cross-attention

formulation (scaled dot-product attention with shared key/value projections) but extends it to the multi-agent sequential-turn setting by replacing the single query with a per-drone query token built from the acting drone’s state.

2.3 Multi-agent reinforcement learning

Cooperative multi-agent reinforcement learning under a shared team reward has matured around three families of methods. Value-decomposition networks [Sunehag et al., 2018] and QMIX [Rashid et al., 2018] assume the team value decomposes additively or monotonically across agents; both scale to large fleets but require a discrete action space and a factored critic. Centralised-training-decentralised-execution proximal policy optimisation, often referred to as MAPPO [Yu et al., 2022], couples a joint-state critic with decentralised actors and is the current strong baseline on many cooperative benchmarks. Attention-augmented multi-agent actors have been explored for communication [Das et al., 2019] and for credit assignment [Iqbal and Sha, 2019].

More recent work applies multi-agent reinforcement learning directly to drone and truck-drone routing. MARL₄DRP [Li et al., 2023] provides the first dedicated benchmarking suite for cooperative drone routing, evaluating value-decomposition, MAPPO, and actor-critic baselines on pickup-and-delivery variants. EMADRL [Zhang et al., 2024] couples an attention encoder with a recurrent decoder and reports consistent gains over flat-MLP baselines on energy-constrained drone fleets. MAGNNET [Lim et al., 2025] pairs a graph-attention-network encoder with a tailored PPO for heterogeneous multi-vehicle task allocation and reaches within 7.5% of a centralised Hungarian optimum while scaling to twenty agents. MHA-PPO [Chen et al., 2025] introduces multi-head attention into the centralised critic for collaborative truck-drone delivery under stochastic demand. Finally, Consensus Multi-Agent Transformer [Hao et al., 2025] uses a Transformer decoder to iterate a shared consensus token that conditions every agent’s action head, reporting strong results on generic cooperative benchmarks.

The present work sits alongside these methods but differs in two respects. First, attention operates over the *environment* (customer set, charger locations) rather than between agents, making the encoder a graph-attention layer over a fully connected node graph. Second, drone-specific behaviour is encoded via a per-drone query token indexed by the acting

drone’s one-hot, so the same encoder is shared across the fleet and the resulting network is permutation invariant across both the customer set and the drone set.

3 Problem Formulation

3.1 Scenario

A fleet of M battery-constrained UAVs shares one common depot, serves N customers with individual deadlines $\{d_i\}_{i=1}^N$, and has access to K en-route chargers. Every drone starts at the depot with full battery $B_{\max}$ and has a constant cruise speed v . Travelling a Euclidean distance δ between nodes consumes $\beta\delta$ units of battery and takes δ/v units of time. A charger visit restores the battery to full and costs a small fixed time τ_c . The global mission horizon is T ; any drone that reaches $B \leq 0$ before arriving at a safe node triggers a crash event with a penalty w_{crash} . Serving customer i at time t pays a service reward w_{serve} minus a tardiness term $w_{\text{tardy}} \max(0, t - d_i)$. Completing the full service set is rewarded with a bonus w_{complete} .

3.2 Sequential-turn multi-agent Markov decision process

The system is formulated as a sequential-turn multi-agent Markov decision process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$ in which exactly one drone acts per global step. The active drone at step t is the drone with the smallest local-elapsed time among the surviving drones,

$$a(t) = \arg \min_{m:\text{alive}} \tau_m(t), \quad (1)$$

which implements an event-driven turn ordering that keeps the global clock monotone non-decreasing. A drone is considered parked (and removed from the action rotation) when it is at the depot and no unvisited customer remains reachable given its current battery and the remaining horizon.

The global state aggregates the node positions, the per-customer visited and deadline status, and the per-drone battery, position, and local clock. The action of the active drone selects a node index from the set $\{0, \dots, D + N + K - 1\}$, where the first D slots are depots (one in the basic scenario), the next N slots are customers, and the last K slots are chargers. A feasibility mask zeros out any action that the active drone cannot reach within

its remaining battery or horizon, or that corresponds to a customer that has already been served. If the mask is empty the fallback action is a return to the depot.

The one-step reward is additive over the drone that just acted and is shared by the whole fleet. Service rewards and tardiness penalties fire only on the step that delivers the customer; the crash penalty fires the step a drone’s battery goes below zero; the completion bonus fires on termination if every customer has been served. Formally,

$$r(t) = w_{\text{serve}} \mathbf{1}\{\text{serve}\} - w_{\text{tardy}} \max(0, t - d_i) \mathbf{1}\{\text{serve}\} - w_{\text{crash}} \mathbf{1}\{\text{crash}\} + w_{\text{complete}} \mathbf{1}\{\text{terminate} \wedge \text{all served}\}. \quad (2)$$

The discount factor is $\gamma = 0.99$, which is close to the undiscounted regime given the short mission horizon but large enough to tighten the credit signal on sparse service rewards.

3.3 Key performance indicators

The evaluation protocol tracks six quantities per episode: team return (sum of per-step rewards), customers served (integer, out of N), crash rate (fraction of episodes with any drone stranded), per-drone utilisation (fraction of time spent serving), load imbalance (coefficient of variation of the per-drone step counts), mean tardiness over served customers, and total distance travelled. The load-imbalance metric is the central measure of coordination quality: a fleet that splits the work evenly has a coefficient of variation near zero, while a fleet that assigns most of the work to one drone has a coefficient of variation that approaches one.

4 Attention-Based Multi-Agent Policy

[To be filled once training completes. Describe the node tokeniser, the multi-layer pre-norm encoder, the per-drone query built from (active one-hot, battery, time, position), the cross-attention scorer, the masked mean-pool team critic, and the proximal policy optimisation training loop. Include Algorithm 1 (attention multi-agent actor-critic rollout) and Algorithm 2 (PPO update with drone-conditioned mini-batches).]



Fig 1 - TBD (architecture)

Figure 1: Attention-based multi-agent policy architecture. (A) Shared node encoder. (B) Per-drone query token. (C) Cross-attention scorer producing action logits. (D) Team critic via masked mean-pool over drone embeddings.

5 Experimental Setup

5.1 Heuristic baselines

Four heuristics cover the main fleet-partitioning archetypes. H1 (cluster + battery-aware nearest-neighbour) partitions the customer set via K-means on customer coordinates, matches clusters to drones with the Hungarian algorithm on centroid-to-drone distance, and runs a battery-aware nearest-neighbour tour inside each cluster. H2 (polar-sweep + battery-aware nearest-neighbour) replaces the K-means partition with an evenly spaced angular slicing around the fleet centroid. H3 (auction + battery-aware nearest-neighbour) runs an iterative auction in which each drone bids its marginal travel cost to append a customer to its tentative tour and the lowest-cost bidder wins. H4 (global earliest-deadline-first dispatch) has no offline partition: at every step the global dispatcher picks, for the currently-acting drone, the unvisited customer with the smallest slack $d_i - (\tau + \text{travel}(m, i))$ among the reachable ones.

5.2 Multi-agent RL baselines

Three multi-agent reinforcement-learning baselines span the standard design space. I-PPO is independent PPO with shared weights across drones; A3C is the advantage actor-critic with sequential-turn rollouts; MAPPO uses a centralised critic over the joint state and a shared actor. The proposed attention-based multi-agent agent (Attn-MARL) replaces the flat-MLP actor and critic of I-PPO and MAPPO with the cross-attention architecture described in Section 4 while keeping the rest of the training pipeline identical.

5.3 Scenario grid

[To be filled.] The main scenario grid spans $M \in \{2, 3, 5\}$, $N \in \{20, 30, 50\}$, two customer spatial distributions (uniform and clustered) and two deadline urgency levels (medium and tight), for a total of 36 scenarios. Each scenario is evaluated with 50 episodes. A separate stress grid adds $M = 10$, $N = 100$, heterogeneous fleets, and multi-depot layouts.

6 Results

[To be filled once experiments complete. Expected figures: (1) learning curves for I-PPO vs. Attn-MARL across the scenario grid, (2) per-method crash-rate bars by fleet size, (3) load-imbalance coefficient-of-variation vs. fleet size, (4) distance-time Pareto frontier with heuristic baselines overlaid, (5) scenario-map example with drone trajectories colored by drone identity.]

7 Discussion

[To be filled once results are available. Cover: scaling behaviour with M and N , the role of permutation invariance on the drone axis, limits of the sequential-turn assumption, and failure modes observed in tight-deadline scenarios.]

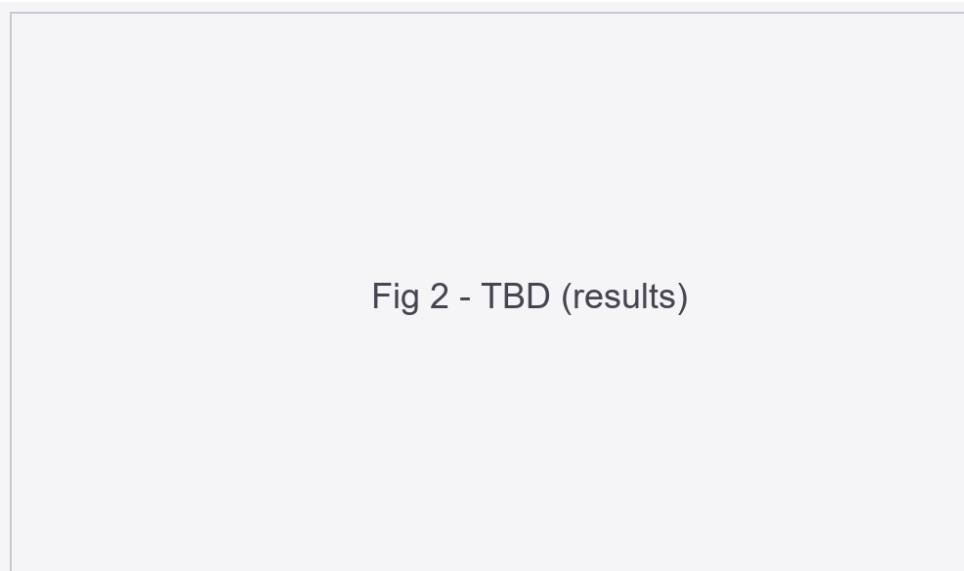


Fig 2 - TBD (results)

Figure 2: Representative results panel. (A) Learning curves (mean team return). (B) Crash rate by fleet size. (C) Load-imbalance coefficient of variation by fleet size. (D) Example fleet trajectories.

8 Conclusion

[To be filled. Recap: sequential-turn MDP, attention-based MARL policy, heuristic and MARL baselines, scaling demonstration. Open problems: communication-augmented variants, heterogeneous fleets, multi-depot dispatch.]

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